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Evgeny Sobolev 09.02.1984 y.b.

$y(x) := \sqrt{x}$ $\quad Max32 := \left(2^{32} - 1\right)$ $\quad clz(x) := \text{if } x = 0 \; 32 \text{ else } 32 - \text{trunc}\left(\log_2\left(\text{trunc}(x) + \frac{Max32}{2^{32}}\right) + \frac{Max32}{2^{32}}\right)$

$msb(x) := 32 - clz(x)$

$dY_32 := \text{trunc}\left(\left(\left(\sqrt{2}\right)^{32} - \left(\sqrt{2}\right)^{31}\right) \cdot 2^{16}\right)$ $\quad Y0_32 := \text{trunc}\left(\sqrt{2} \cdot 2^{15} \cdot 2^{16}\right)$

$dY_31 := \text{trunc}\left(\left(\left(\sqrt{2}\right)^{31} - \left(\sqrt{2}\right)^{30}\right) \cdot 2^{16}\right)$ $\quad Y0_31 := \text{trunc}\left(2^{15} \cdot 2^{16}\right)$

$dY_32 = 1257966796$ $\quad dY_31 = 889516851$

$Y0_32 = 3037000499$ $\quad Y0_31 = 2147483648$

$dY(x) := \text{if } \text{trunc}\left(\frac{clz(x)}{2}\right) \cdot 2 = clz(x) \; dY_32 \text{ else } dY_31$

$Y0(x) := \text{if } \text{trunc}\left(\frac{clz(x)}{2}\right) \cdot 2 = clz(x) \; Y0_32 \text{ else } Y0_31$

$xScaled(x) := \text{trunc}\left(\text{trunc}(x) \cdot 2^{clz(x)+1} - 2^{32}\right)$

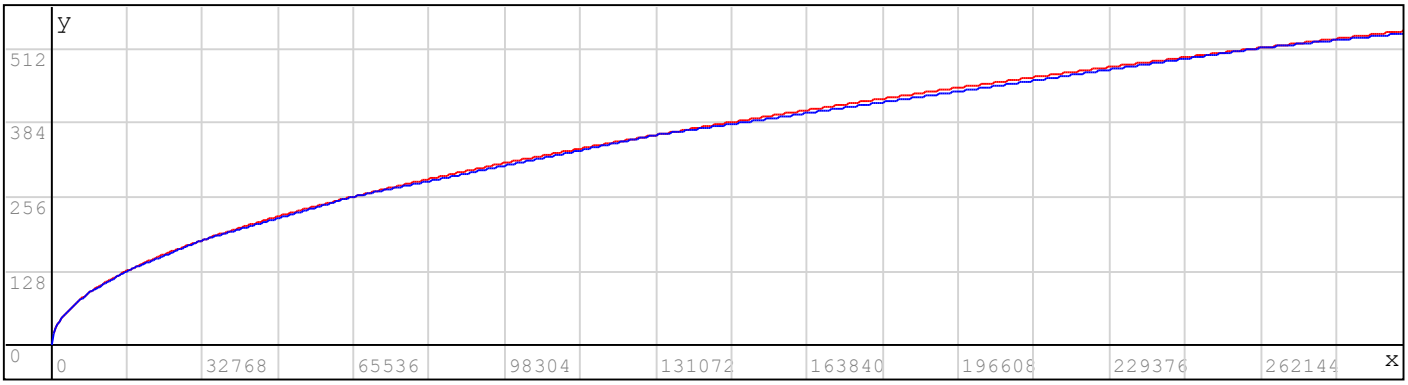
$sqrtLinearScaled(x) := \text{trunc}\left(\frac{\text{trunc}(xScaled(x) \cdot dY(x))}{2^{32}}\right) + Y0(x)$

$sqrtLinear(x) := \frac{sqrtLinearScaled(x)}{2^{\text{trunc}\left(\frac{clz(x)}{2}\right)}}$

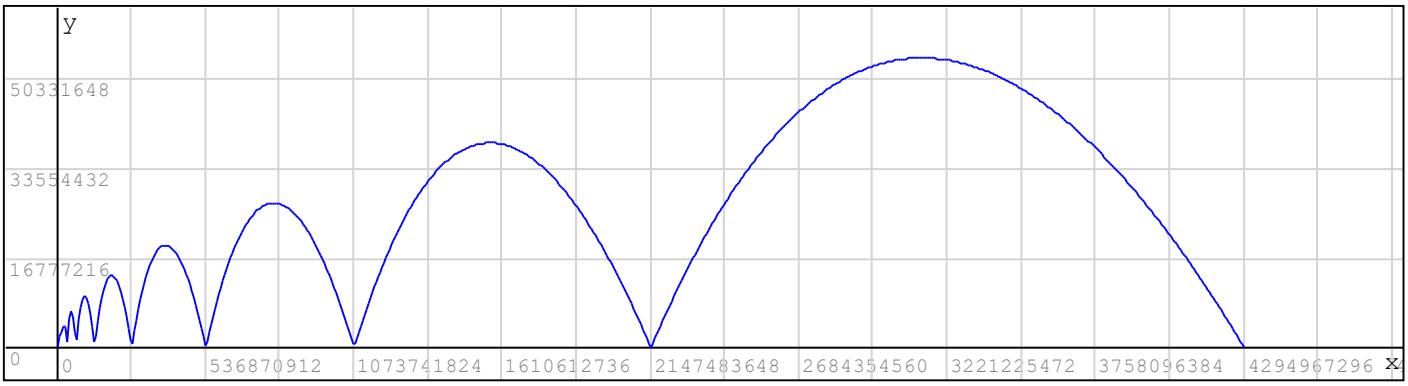
$sqrtLinearErr(x) := y(x) \cdot 2^{16} - sqrtLinear(x)$

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$$\begin{cases} \frac{sqrtLinear(x)}{2^{16}} \\ \sqrt{x} \end{cases}$$



$$\left\{ \sqrt{x} \cdot 2^{16} - sqrtLinear(x) \right\}$$

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$$xMaxErr_32 := \operatorname{trunc}\left(\frac{2^{32}}{\left(4-2\cdot\sqrt{2}\right)^{2}}\right) \qquad xMaxErrScaled_32 := \operatorname{trunc}\left(\frac{2\cdot\sqrt{2}-1}{4}\cdot 2^{32}\right)$$

$$yMaxErr_31 := \operatorname{trunc}\left(\sqrt{\frac{xMaxErr_32}{2}}\cdot 2^{16} - sqrtLinear\left(\frac{xMaxErr_32}{2}\right)\right)$$

$$yMaxErr_32 := \operatorname{trunc}\left(\sqrt{xMaxErr_32}\cdot 2^{16} - sqrtLinear\left(xMaxErr_32\right)\right)$$

$$xMaxErr_32 = 3129112985 \qquad yMaxErr_31 = 38154242 \qquad yMaxErr_32 = 53958246$$

$$xMaxErrScaled_32 = 1963258675$$

$$yMaxErrScaled(x) := \text{if } \operatorname{trunc}\left(\frac{clz(x)}{2}\right)\cdot 2 = clz(x) \quad yMaxErr_32 \text{ else } yMaxErr_31 \qquad \frac{yMaxErr_32}{65536} = 823.33749$$

$$xMaxErr(x) := \frac{xMaxErr_32}{2^{\operatorname{trunc}(clz(x))}} \qquad yMaxErr(x) := \frac{yMaxErrScaled(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}}$$

$$dXMaxErr(x) := \left|\left(xScaled(x) - xMaxErrScaled_32\right)\cdot 2\right|$$

$$dXMaxErr(x) := \text{if } xScaled(x) < xMaxErrScaled_32 \left|\left(xMaxErrScaled_32 - xScaled(x)\right)\right|\cdot\frac{Max32}{xMaxErrScaled_32} \text{ else } \left|\left(xScaled(x) - xMaxErrScaled_32\right)\right|\cdot\frac{Max32}{Max32 - xMaxErrScaled_32}$$

$$dQMaxErr(x) := \operatorname{trunc}\left(\frac{\left(dXMaxErr(x)\right)^{2}}{2^{32}}\right) \qquad dXMidErr(x) := \left|2^{31} - xScaled(x)\right|\cdot 2$$

$$dQMaxErrInv(x) := 2^{32} - dQMaxErr(x) \qquad dQMidErr(x) := \operatorname{trunc}\left(\frac{\left(dXMidErr(x)\right)^{2}}{2^{32}}\right)$$

$$parMaxErr(x) := \operatorname{trunc}\left(\frac{dQMaxErrInv(x)\cdot yMaxErrScaled(x)}{2^{32}}\right) \qquad dQMidErrInv(x) := 2^{32} - dQMidErr(x)$$

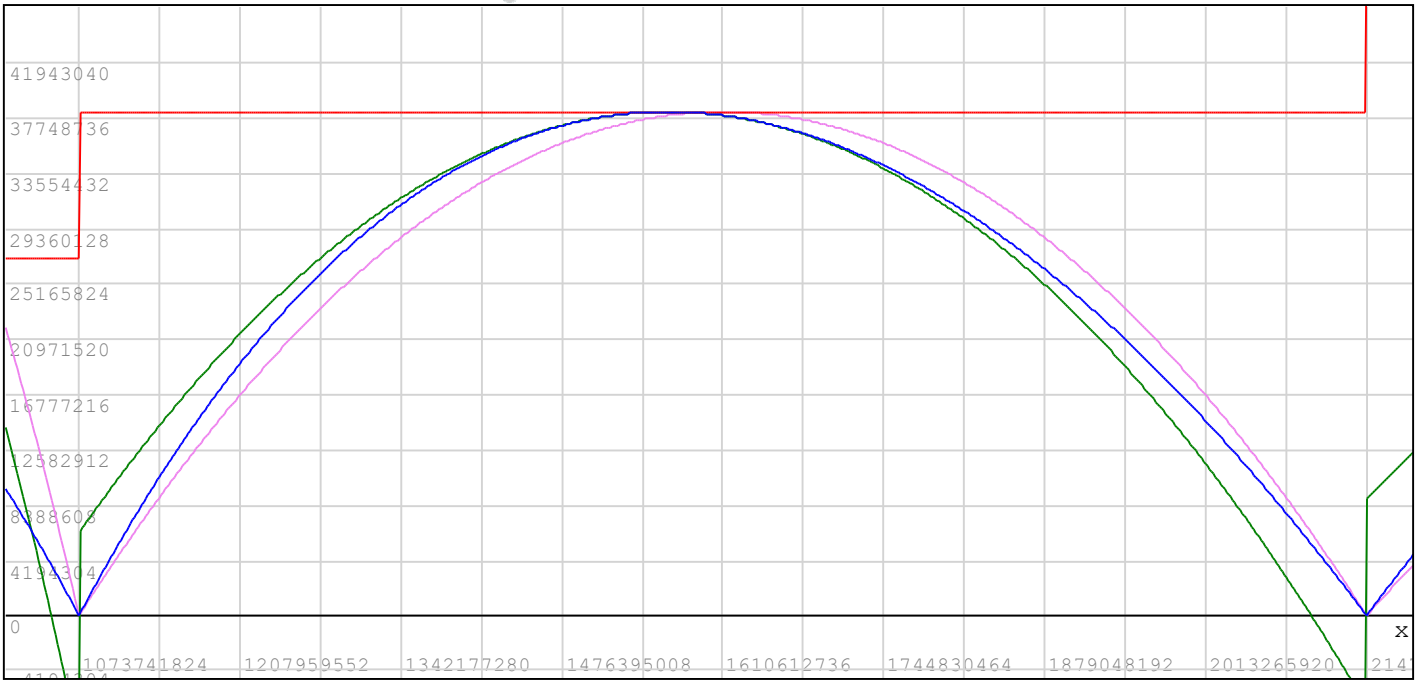
$$Kmix := 0.03648649$$

$$parMidErr(x) := \operatorname{trunc}\left(\frac{dQMidErrInv(x)\cdot yMaxErrScaled(x)}{2^{32}}\right) \qquad \operatorname{trunc}\left(Kmix\cdot 2^{32}\right) = 156708281$$

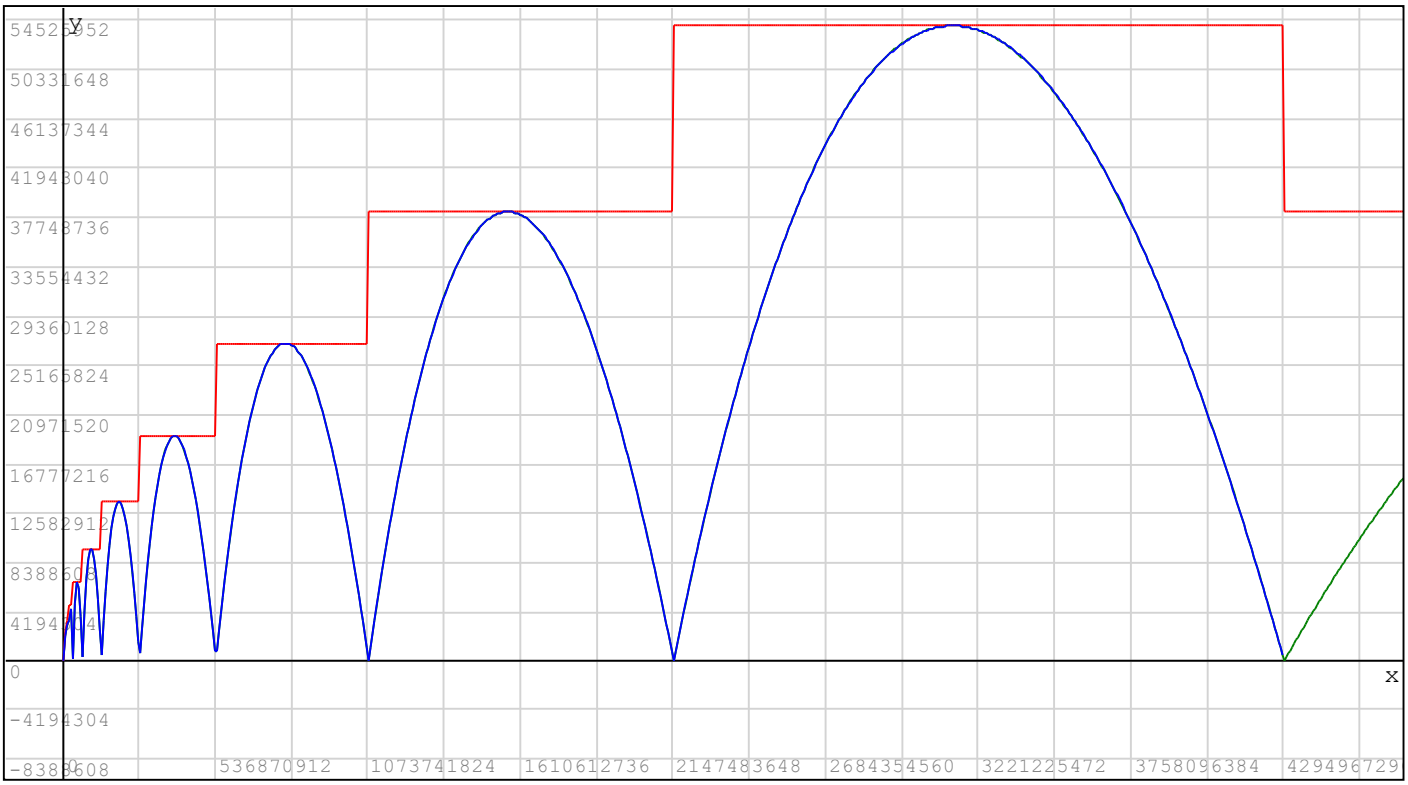
$$par(x) := \left(\operatorname{trunc}\left(\frac{parMaxErr(x)\cdot dQMidErrInv(x)}{2^{32}}\right) + \operatorname{trunc}\left(\frac{parMidErr(x)\cdot dQMidErr(x)}{2^{32}}\cdot\left(1+\frac{\operatorname{trunc}(156708281)}{2^{32}}\right)\right)\right.$$

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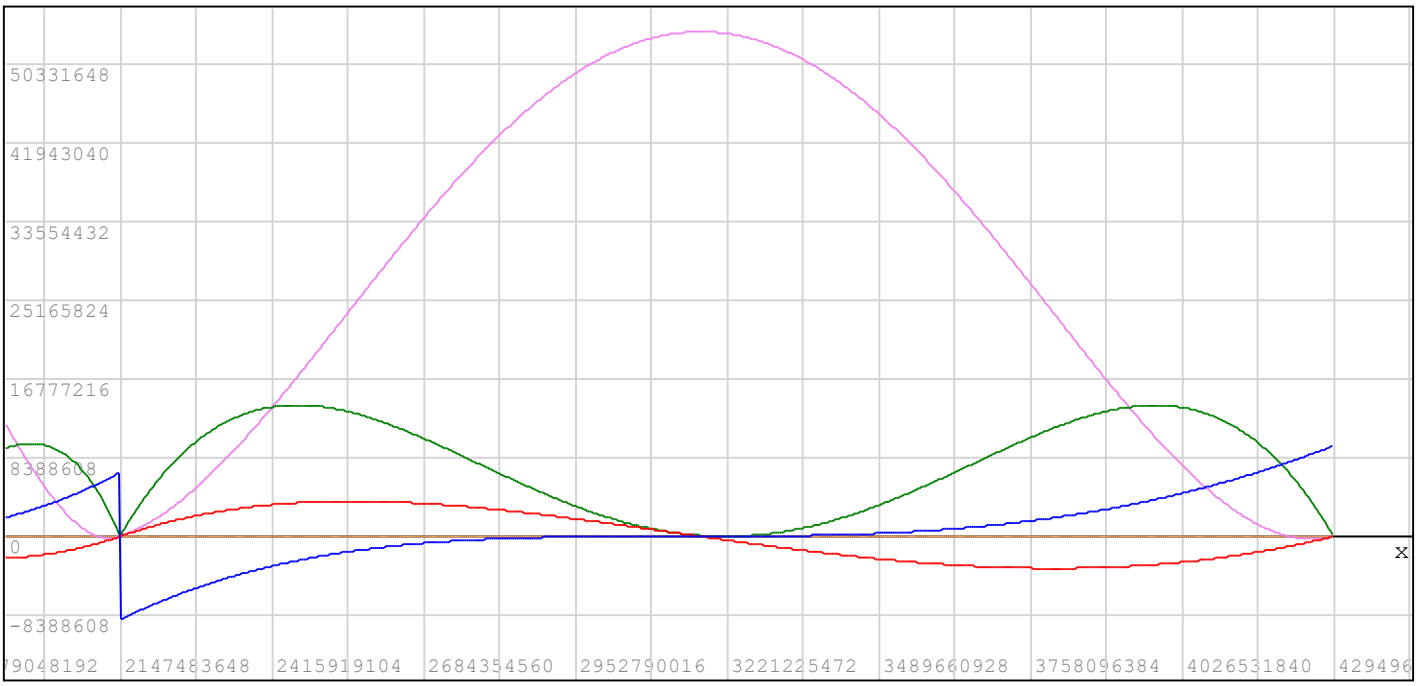
$$\begin{cases} sqrtLinearErr(x) \\ yMaxErr(x) \\ parMaxErr(x) \\ parMidErr(x) \end{cases}$$



$$\begin{cases} sqrtLinearErr(x) \\ yMaxErr(x) \\ \dfrac{par(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}} \end{cases}$$

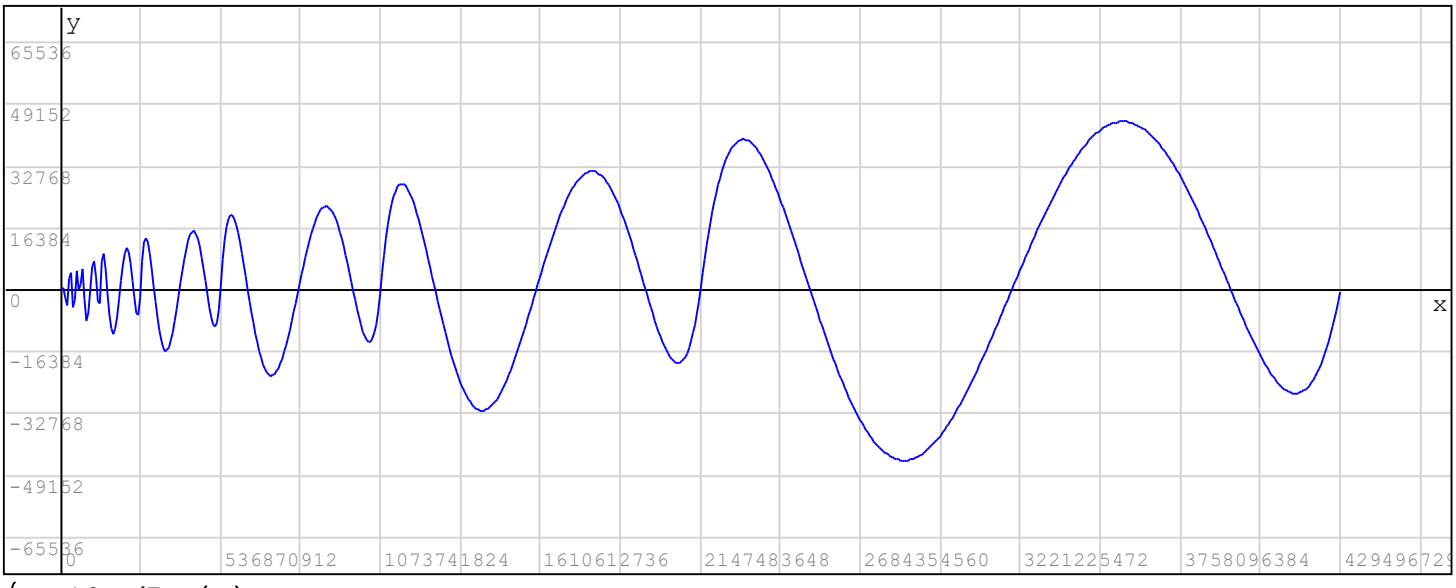
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$$\begin{cases} sqrtLinearErr(x) - parMaxErr(x) \\ sqrtLinearErr(x) - parMidErr(x) \\ sqrtLinearErr(x) - parMaxErr(x) \cdot \frac{dQMidErrInv(x)}{2^{32}} \\ sqrtLinearErr(x) - parMidErr(x) \cdot \frac{dQMidErr(x)}{2^{32}} \\ sqrtLinearErr(x) - par(x) \end{cases}$$

$$sqrtQuadErr(x) := y(x) \cdot 2^{16} - \frac{sqrtLinearScaled(x) + par(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}}$$



$$\{sqrtQuadErr(x)$$

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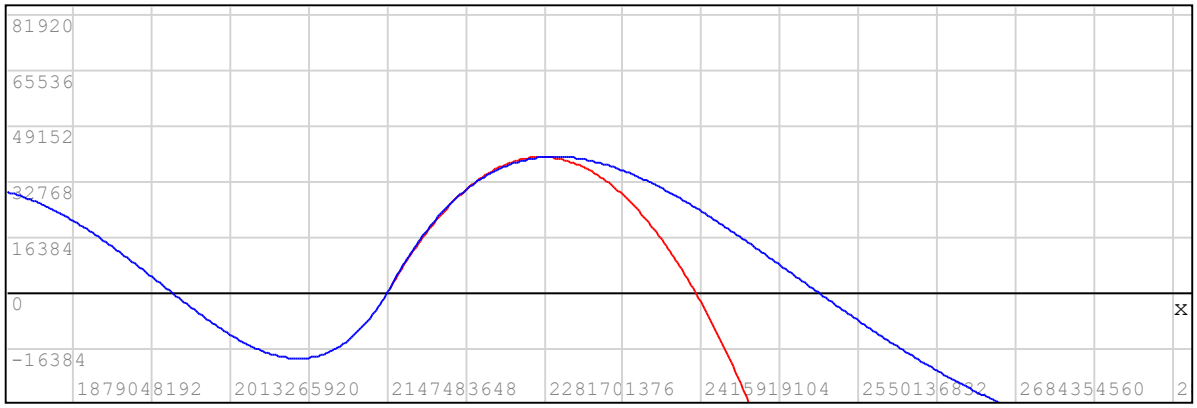
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$$xErr1(x) := \text{trunc}\left(\left|xScaled(x) - 2^{31-3}\right|\right) \cdot 2^{4}$$

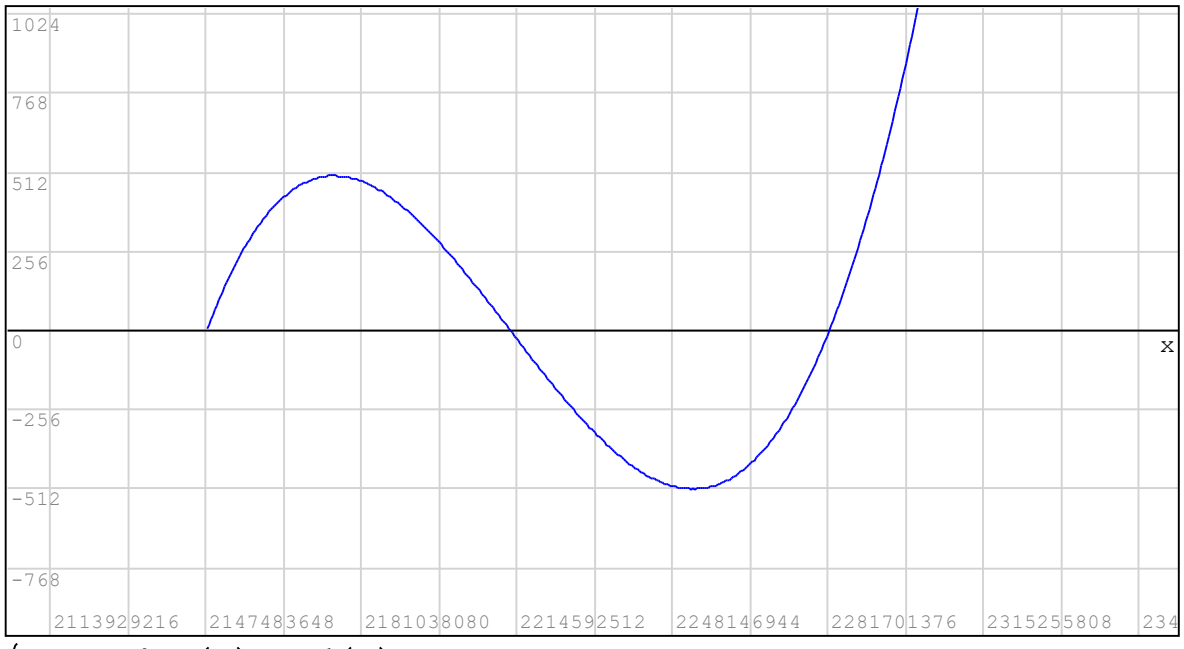
$$E1a := 0.03248 \qquad E1b := \left(\frac{\text{trunc}\left(sqrtQuadErr\left(2^{31} + 2^{31-4}\right)\right)}{44528} + E1a\right)$$

$$E1a_t := \text{trunc}\left(E1a \cdot 2^{32}\right) = 139500537 \qquad E1b_t := \text{trunc}\left(E1b \cdot 2^{32}\right) = 3978426570$$

$$err1(x) := \text{trunc}\left(\frac{\left(E1b_t - \text{trunc}\left(\frac{\text{trunc}\left(\frac{xErr1(x) \cdot xErr1(x)}{2^{32}}\right)}{2^{32}} \cdot E1b_t\right) - \text{trunc}\left(\frac{\text{trunc}\left(xScaled(x) \cdot 2^{4} \cdot E1a_t\right)}{2^{32}}\right)\right)}{2^{32}}\right.$$



$$\begin{cases} sqrtQuadErr(x) \\ \dfrac{err1(x)}{2^{\frac{clz(x)}{2}}} \end{cases}$$



$$\{sqrtQuadErr(x) - err1(x)$$

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$$xErr2(x) := xErr1(x)$$

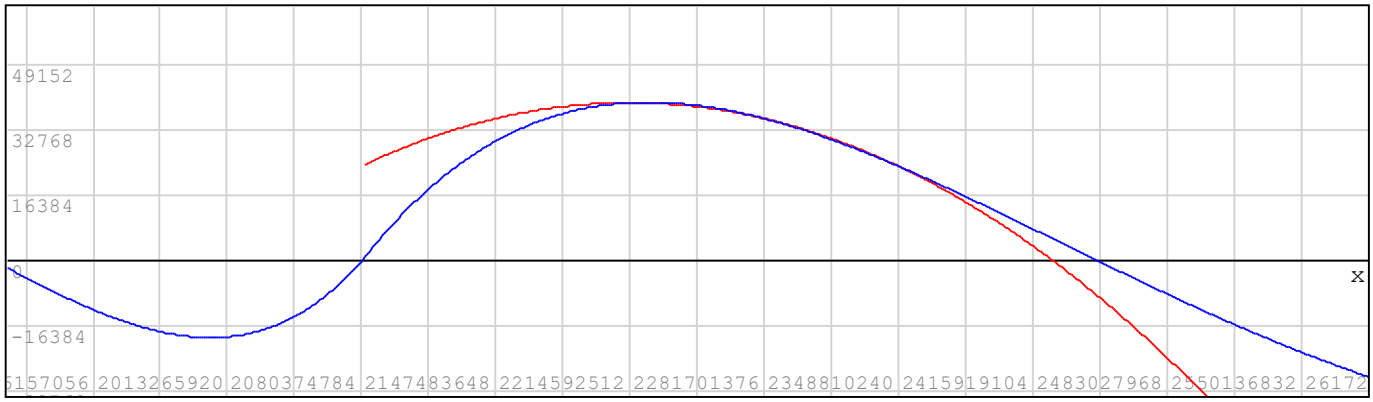
$$E2a := -0.0035 \qquad E2b := 0.36 \qquad E2c := \frac{\text{trunc}\left(sqrtQuadErr\left(2^{31} + 2^{31-4}\right)\right)}{44528}$$

$$E2a_t := \text{trunc}\left(E2a \cdot 2^{32}\right) = -15032385$$

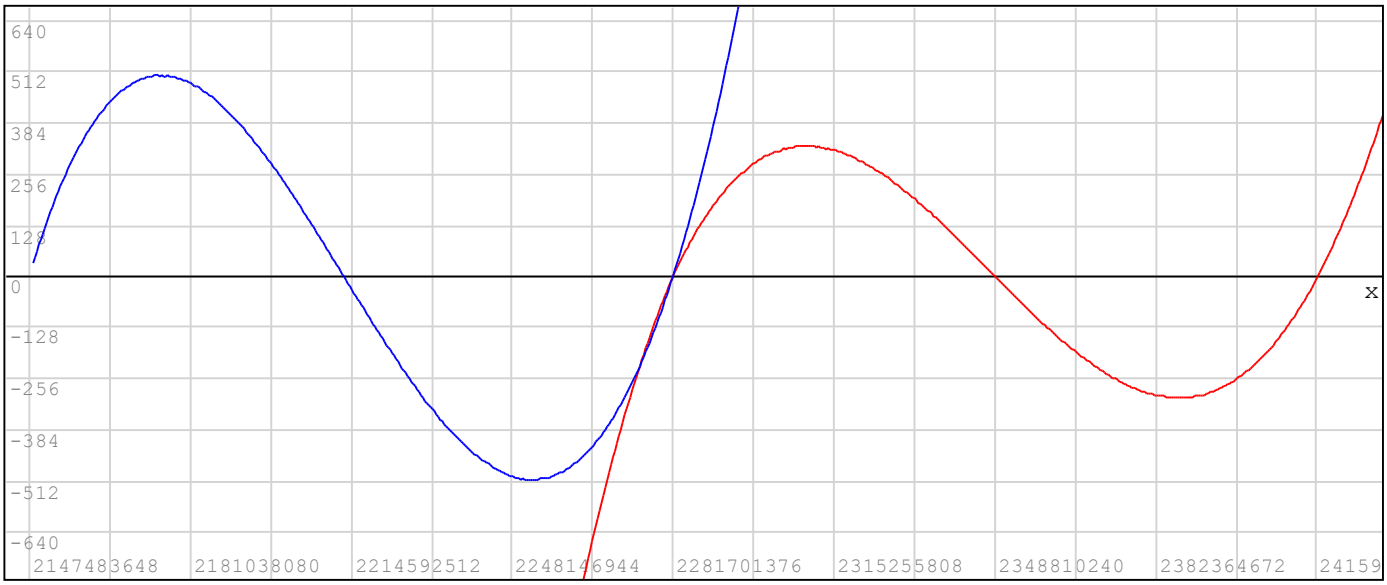
$$E2c_t := \text{trunc}\left(E2c \cdot 2^{32}\right) = 3838926032$$

$$E2b_t := \text{trunc}\left(E2b \cdot 2^{32}\right) = 1546188226$$

$$err2(x) := \text{trunc}\left(\frac{\left[E2c_t - \text{trunc}\left(\frac{\text{trunc}\left(\frac{xErr2(x) \cdot xErr2(x)}{2^{32}}\right)}{2^{32}} \cdot E2b_t\right) - \text{trunc}\left(\frac{\text{trunc}\left(xErr2(x) \cdot E2a_t\right)}{2^{32}}\right)\right] \cdot 44528}{2^{32}}\right)$$



$$\begin{cases} sqrtQuadErr(x) \\ \dfrac{err2(x)}{2^{\frac{clz(x)}{2}}} \end{cases}$$



$$\begin{cases} sqrtQuadErr(x) - err1(x) \\ sqrtQuadErr(x) - err2(x) \end{cases}$$

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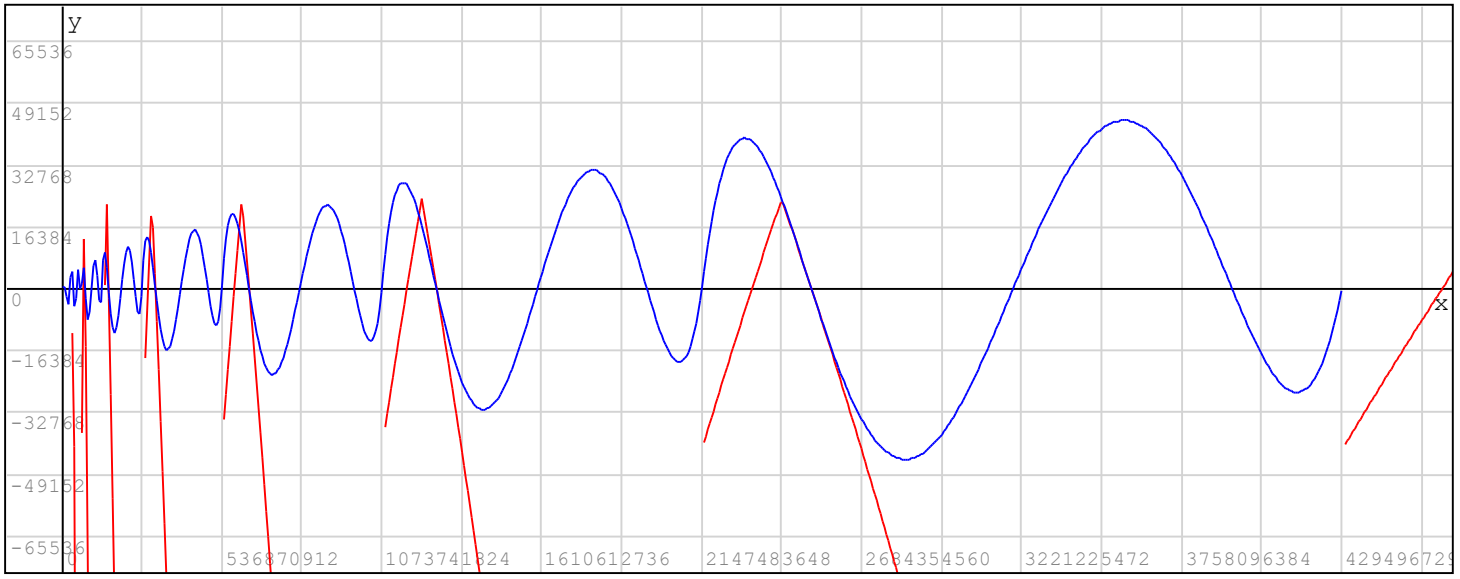
$$xErr3(x) := \text{trunc}\left(\left|xScaled(x) - 2\cdot 2^{31-3}\right|\right)\cdot 2^{4}$$

$$E3a := 0.695 \qquad E3b := 0.0235 \qquad E3c := \frac{sqrtQuadErr\left(2^{31} + 2^{31-3}\right)}{44528}$$

$$E3a_t := \text{trunc}\left(E3a\cdot 2^{32}\right) = 2985002270 \qquad E3c_t := \text{trunc}\left(E3c\cdot 2^{32}\right) = 2307117381$$

$$E3b_t := \text{trunc}\left(E3b\cdot 2^{32}\right) = 100931731$$

$$err3(x) := \text{trunc}\left(\frac{\left(\left(E3c_t - \text{trunc}\left(\frac{\text{trunc}\left(\frac{xErr3(x)\cdot xErr3(x)}{2^{32}}\right)}{2^{32}}\cdot E3b_t\right) - \text{trunc}\left(\frac{\text{trunc}\left(xErr3(x)\cdot E3a_t\right)}{2^{32}}\right)\right)\cdot 4452\right.}{2^{32}}\right.$$



$$\begin{cases} sqrtQuadErr(x) \\ err3(x) \end{cases}$$

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$$\begin{cases} sqrtQuadErr(x) - err1(x) \\ sqrtQuadErr(x) - err2(x) \\ sqrtQuadErr(x) - err3(x) \end{cases}$$

$$xErr4(x) := \mathrm{trunc}\left(\left|xScaled(x) - 4 \cdot 2^{31-3}\right|\right) \cdot 2^{4}$$

$$E4a := 0.4418 \qquad E4c := -\frac{sqrtQuadErr\left(2^{31} + 2 \cdot \left(2^{31-3}\right)\right)}{44528}$$

$$E4b := 0.584 - E4a$$

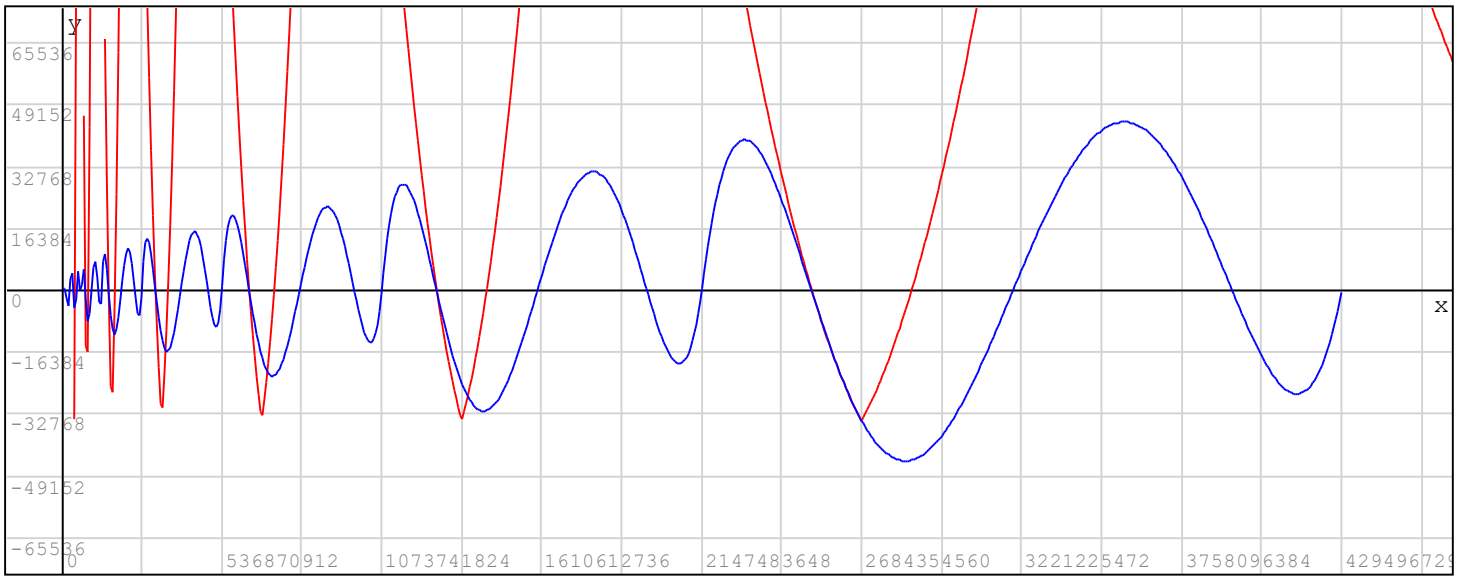
$$E4a_t := \mathrm{trunc}\left(E4a \cdot 2^{32}\right) = 1897516551 \qquad E4c_t := \mathrm{trunc}\left(E4c \cdot 2^{32}\right) = 3287314450$$

$$E4b_t := \mathrm{trunc}\left(E4b \cdot 2^{32}\right) = 610744349 \qquad E4c = 0.7654$$

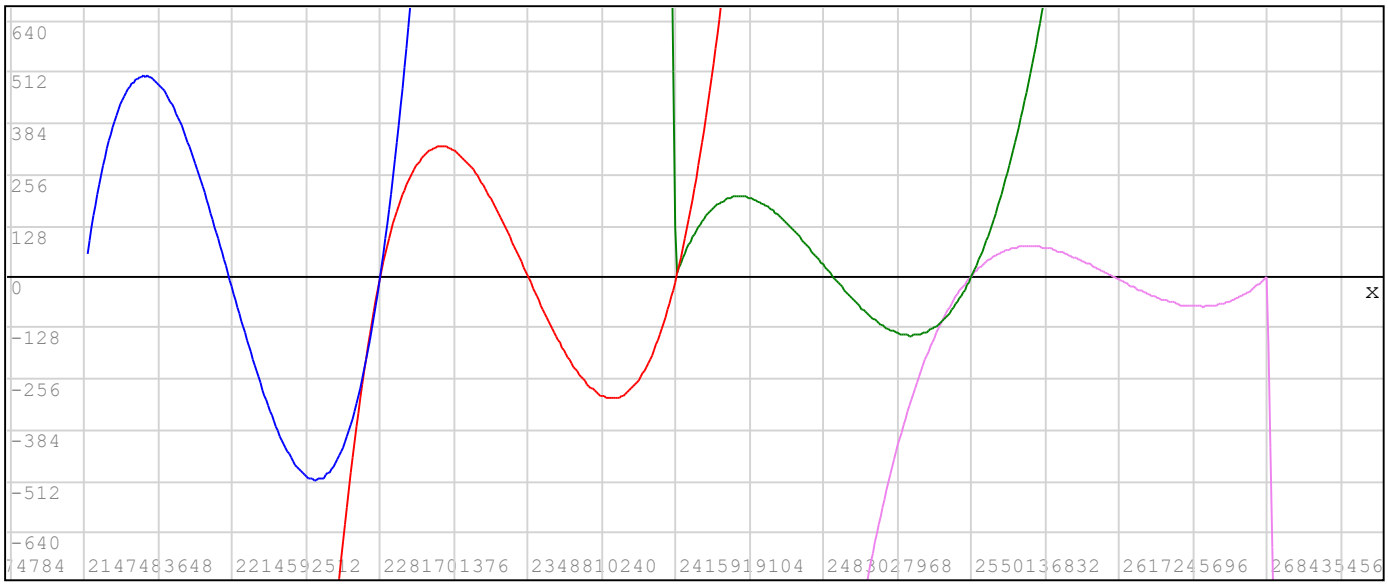
$$err4(x) := \mathrm{trunc}\left(\frac{\left(\mathrm{trunc}\left(\frac{\mathrm{trunc}\left(\frac{xErr4(x) \cdot xErr4(x)}{2^{32}}\right)}{2^{32}} \cdot E4b_t\right) + \mathrm{trunc}\left(\frac{\mathrm{trunc}\left(xErr4(x) \cdot E4a_t\right)}{2^{32}}\right) - E4c_t\right) \cdot 44528}{2^{32}}\right.$$

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$$\begin{cases} sqrtQuadErr(x) \\ err4(x) \end{cases}$$



$$\begin{cases} sqrtQuadErr(x) - err1(x) \\ sqrtQuadErr(x) - err2(x) \\ sqrtQuadErr(x) - err3(x) \\ sqrtQuadErr(x) - err4(x) \end{cases}$$

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